

Which chart? A decision card.

Pick the chart that answers your question — not the one that looks impressive.

Step 1 — Name your question, then read across

My question is...	Use this	Not this	Why
How many / how much in each category ?	Bar chart	Pie chart	The eye compares lengths accurately and angles badly
How are the values of one continuous variable spread out?	Histogram	Bar chart	A histogram has no gaps — the data are continuous
How does a variable change over time ?	Line chart	Bar chart	A line shows trend and direction; bars fight it
How do groups compare on a distribution ?	Box plot	Bar of means	A bar of means hides spread, skew and outliers completely
Is there a relationship between two continuous variables?	Scatter plot	Line chart	A line implies a sequence that does not exist
What proportion of a whole?	Stacked bar (100%)	Pie chart	Several stacked bars can be compared side by side; several pies cannot
Where does the cohort lose people ?	Flow diagram / funnel	Table	Attrition is the story — make it visible

Step 2 — Run the five checks before you paste it into your report

- 1. Do both axes have labels, with units?** “Age (years)”, not “age”. This is the single most common omission.
- 2. Does the y-axis start at zero?** For bar charts it *must*. Truncating the axis exaggerates small differences and is the most common way graphs mislead — sometimes on purpose.
- 3. Is n stated?** Put it in the title or the axis label: “(n = 160)”.
- 4. Would it survive being photocopied in black and white?** Many reports are. Do not rely on colour alone to carry meaning — use patterns, labels or shading too.
- 5. Have you removed the decoration?** No 3-D, no drop shadows, no gradient fills, no gridlines you do not need, no legend when a direct label on the line would do.

The four ways a graph lies

1. The truncated axis

Starting the y-axis at 65% instead of 0% can turn a 3-point difference into a cliff. If you must zoom in, say so clearly on the chart.

2. The missing denominator

A bar showing '15 cases in Covalima' next to '57 in Dili' says nothing until you know Dili has six times the population. Plot **rates**, not counts, when comparing places.

3. The pie chart with nine slices

Nobody can rank nine angles. If you have more than about four categories, use a bar chart sorted from largest to smallest.

4. The bar chart of means

Two groups can have identical mean weights while one is tightly clustered and the other is wildly spread. A bar of means hides this completely. Use a box plot.

TIP

A table or a graph — which? Use a **table** when the reader needs exact values, or when you have many variables to report. Use a **graph** when the shape, the trend or the comparison is the message. Never show the same numbers as both — pick one, and put the other in an annex if a reviewer asks.

Liafuan importante • Key words in Tetum

Graph, chart	Grafiku
Bar chart	Grafiku barra
Line	Liña
Compare	Kompara
Increase / decrease	Aumenta / hamenus
Picture, image	Imajen